New upper bounds for strength-3 covering arrays

October 2026. The reference bounds are the Colbourn covering array tables (November 2024 status), as preserved by U. Grömping in the GitHub repository `ugroempi/CAs`*. The original tables have not been maintained since 2025.*

The problem

A covering array CA(N; t, k, v) is an N × k array over v symbols in which every N × t subarray contains every t-tuple of symbols at least once. In software testing, the rows are test cases: k parameters with v values each, such that every combination of t parameter values is tested. CAN(t, k, v), the smallest possible N, is unknown for almost all parameters, so the community keeps tables of the best known upper bounds.

Results: 78 improved entries

v	k	New N	Previous bound (Nov 2024)	Gain
5	32	**324**	349	−25
5	33–36	**325**	349	−24
5	37	**350**	365	−15
7	58	**930**	979	−49
7	59–64	**931**	1014	−83
7	65	**980**	1015	−35
8	74	**1407**	1447	−40
8	75	**1408**	1447	−39
8	76–81	**1408**	1455	−47
9	92	**2024**	2105	−81
9	93–100	**2025**	2169	−144
9	101	**2106**	2169	−63
11	134	**3750**	3871	−121
11	135–144	**3751**	3971	−220
11	145	**3872**	3971	−99
13	184	**6252**	6421	−169
13	185–196	**6253**	6565	−312
13	197	**6422**	6565	−143
19	382	**19854**	20215	−361
19	383–400	**19855**	20539 to 20575	−684 to −720
19	401	**20216**	20575	−359

The full list is in `improvements.csv`. The construction works for every prime power $q \geq 3$, but gives no improvement for $q = 3, 4, 16, 17, 23$ or 25 (the tables stop at $v = 25$).

Construction

For a prime power $q \geq 3$ and $m \geq 2$ added columns ($m \leq q^2 + q + 1$):

> **CA($2q^3 + q^2 \cdot L$; 3, $q^2 + q + 1 + m$, q)**, with $L = \max(q - 2, \lceil m(q-2)/q \rceil)$

For $m = 1$: CA($2q^3 - 1 + q^2(q-2)$; 3, $q^2 + q + 2$, q).

All arithmetic is in GF(q). The elements other than 0 and 1 are denoted $e_2, \ldots, e_{q-1}$ (in the code: the integers $2, \ldots, q-1$).

1. **Base (Raaphorst–Moura–Stevens, 2014)**: CA($2q^3 - 1$; 3, $q^2 + q + 1$, q).
   - Columns $j = 0, \ldots, k_0 - 1$ ($k_0 = q^2 + q + 1$) are the points $p_j = \alpha^j$ of PG(2, q), where α is a root of a primitive cubic over GF(q) (Singer cycle).
   - Block B1 has one row per $x \neq 0$ in GF(q)^3, with entries $\langle x, p_j \rangle$.
   - Block B2 is the same with the column order reversed (column j uses p_{k_0-1-j}).
   - There is one all-zero row.
   - RMS theorem: three columns whose points are collinear in B1 have non-collinear points in B2.
2. **Shifted copies**: new column n_i ($i = 1, \ldots, m$) is a copy of column $c_i = i - 1$ on the zero row and on B1, and equals **$c_i + 1$** on B2.
3. **Layers** ($m \geq 2$): L layers of q^2 rows, indexed by $y \in$ GF(q)^2.
   - The $q + 1$ directions $P_0 = (1,0)$, $P_1 = (0,1)$, $P_{1+j} = (1, a_j)$ ($a_j \neq 0$) of PG(1, q) give linear forms $y \mapsto y \cdot P_s$, any two of which form a bijection GF(q)$^2 \to$ GF(q)^2.
   - The $m(q-2)$ "activations" (i, e), with $e \in \{e_2, \ldots, e_{q-1}\}$, are distributed over the layers so that each layer contains at most q activations, with distinct copies. If $m \leq q$, layer ℓ activates every copy with $e = e_{\ell+1}$. If $m > q$, the list of activations, ordered by e then i, is cut into consecutive windows of q; the copies in a window are distinct because $m > q$.
   - In a layer, the s-th active copy ($s = 1, \ldots,$ at most q) with value e gets $c_i = y \cdot P_s$ and $n_i = y \cdot P_s + e$. Every other column (original or new) gets $y \cdot P_0$.
4. **One correction row** ($m \geq 2$): 0 on the original columns, 1 on the new columns.
5. **For $m = 1$**, the layers and the correction row are replaced by the $q^2(q-2)$ rows ($c_1 = a$, $n_1 = a + e$, every other column $= s$), for $a, s \in$ GF(q) and $e \in \{e_2, \ldots, e_{q-1}\}$.

Proof

Fix three distinct columns T. We show that every 3-tuple of values appears on T.

Case 1: T has three original columns. By the RMS theorem, the three points are non-collinear in B1 or in B2. In that block, $x \mapsto (\langle x, p_a \rangle, \langle x, p_b \rangle, \langle x, p_c \rangle)$ is a bijection of GF(q)^3. The non-zero x give every tuple except $(0, 0, 0)$, which is on the zero row.

Case 2: T contains new columns, but no pair $\{c_i, n_i\}$.

- Replace each new column n_i by its source c_i. This gives three distinct original columns (two new columns have distinct sources), and therefore three points that are non-collinear in B1 or in B2.
- On B1, the values on T are exactly those of the sources. On B2, they are those of the sources plus a constant vector δ (1 on the new columns, 0 on the others).
- So on the good block, x ↦ (values on T) is a bijection of $GF(q)^3$ followed by a translation. The non-zero x give every tuple except the image of x = 0:
  - on B1, the image of 0 is (0, 0, 0), which is on the zero row (new columns copy their sources there);
  - on B2, it is δ, which is on the correction row for m ≥ 2.
  - For m = 1, δ has a single 1 on n_1, and $c_1 \notin T$. Choose $a \notin \{0, 1\}$ and $e = 1 - a$ (then $e \notin \{0, 1\}$). The added row with $c_1 = a$, $n_1 = 1$ and s = 0 covers δ.

Case 3: T = {c_i, n_i, w}. Write $\varepsilon = n_i - c_i$. It suffices to show that, for each $\varepsilon \in GF(q)$, the pair (c_i, w) takes all q^2 values on the rows where $n_i - c_i = \varepsilon$.

- **ε = 0, on B1.** The sources of c_i and w are distinct points. So $x \mapsto (\langle x, p\rangle, \langle x, p'\rangle)$ is a linear surjection $GF(q)^3 \to GF(q)^2$, with a one-dimensional kernel. Every pair is the image of q vectors x, and at least $q - 1 \geq 1$ of them are non-zero. This also holds if w is another new column n_j.
- **ε = 1, on B2.** Same argument. The constant shift of n_j does not change surjectivity.
- **$\varepsilon = e \notin \{0, 1\}$, m ≥ 2.** Take the layer where (i, e) is active, with direction P_s. Then w carries a different direction P_{s′}, plus possibly a constant: P_0 if w is not active in that layer, or the direction of its own activation otherwise. Hence y ↦ (c_i, w) is a bijection of $GF(q)^2$ followed by a translation, so every pair appears.
- **$\varepsilon = e \notin \{0, 1\}$, m = 1.** The rows ($c_1 = a$, $n_1 = a + e$, w = s) give all pairs (a, s).

∎

Run count. The base has $1 + 2(q^3 - 1)$ rows. The layers add q^2L rows, with $L = \max(q - 2, \lceil m(q - 2)/q \rceil)$, and the correction row adds 1, for a total of $2q^3 + q^2L$. For m = 1, the added rows are $q^2(q - 2)$.

Verification

Every array in `arrays/` was checked by brute force (every triple of columns, every value combination) with checkers written independently of the construction code: `verify.py` (Python) and `verify.c` (C).

- Sanity check of the checkers: removing one row of CA(1407;3,74,8), or changing a single cell of CA(2025;3,100,9), is correctly reported as INVALID.
- The arrays for q = 19 (about 20 MB each) and a CA(11776; 3, 289, 16) were also verified but are not included. Run `python construct_layers.py 20 19` to regenerate them.
- The R implementation in `R-contribution/` produces the same arrays, up to row order, for q = 4, 5, 7, 8, 9.
- For prime q, the GF(q) version of the code produces arrays byte-identical to the earlier prime-only version.

Files

- `gf.py` : GF(q) arithmetic for prime powers. The field axioms are checked exhaustively when it is built.
- `rms_base.py` : rebuilds the RMS base array.
- `construct_one_copy.py` : construction with one added column (m = 1).
- `construct_layers.py` : general construction (m ≥ 2).
- `verify.py` , `verify.c` : independent checkers.
- `arrays/CA_N_t_k_v.txt` : the arrays, one test per line, values 0 ... v − 1.
- `improvements.csv` : the 78 improved entries.
- `search/` : exploration tools used along the way (simulated annealing in C, CP-SAT model).
- `R-contribution/` : an implementation for the R package `CAs` , with the corresponding git patch:
  - functions `RMScopies` , `RMScopiesCA` and `N_RMScopiesCA` , for prime powers via `lhs::create_galois_field` ;
  - documentation and tests.

Caveats

- Novelty is established **relative to the November 2024 tables** and to the published constructions I could find (Shokri, Moura and Stevens 2023–2025 start at k = 2(q² + q + 1)). A more recent unindexed result is possible, and the idea may be considered folklore by specialists.
- The proof above has not yet been checked by an independent human expert.

Acknowledgement

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